

YAMBA, Australia

WMO: 945890

Lat: 29.4333S

Lon: 153.3633E

Elev: 28

StdP: 100.99

Time Zone: 10.00 (AUS)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	9.1	-0.2	3.7	14.6	1.9	4.3	14.1	11.1	18.0	9.1	18.0	3.3	240	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.6	29.3	23.9	28.2	23.4	27.4	22.9	25.6	27.3	24.8	26.8	24.2	26.1	4.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.3	26.6	24.3	19.3	25.8	23.6	18.5	25.3	79.0	27.1	75.9	26.9	73.2	26.1	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.1	8.0	6.2	33.7	0.6	2.3	5.7	35.3	5.4	36.7	5.0	38.0	4.6	39.6		
			3.8	26.2	0.7	1.6	3.4	27.3	3.0	28.2	2.6	29.1	2.1	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	24.1	24.0	23.3	20.9	17.8	15.7	15.0	15.8	18.1	19.8	21.6	22.9
	DBStd	3.69	1.43	1.51	1.48	1.72	1.90	1.85	1.99	2.17	2.15	2.08	1.72	1.55
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	341	0	0	0	2	32	79	104	84	29	10	1	0
	CDD10.0	3613	436	393	413	326	242	172	156	180	244	303	348	399
	CDD18.3	912	178	160	154	78	16	1	2	6	24	54	98	141
	CDH23.3	3782	943	886	675	197	35	1	5	32	60	139	273	536
Wind	WSAvg	3.9	4.1	4.5	4.1	4.0	3.4	3.6	3.4	3.7	3.8	4.1	4.1	4.0
	PrecAvg	1544	134	172	204	175	182	138	87	62	49	85	102	130
	PrecMax	2438	479	558	618	629	554	548	300	184	214	371	251	385
	PrecMin	825	13	38	41	31	10	12	2	2	0	6	6	38
	PrecStd	388	95	102	134	126	129	98	75	50	47	68	59	81
	DB	29.8	30.3	29.9	28.2	26.1	22.8	23.8	26.6	27.1	28.3	30.4	29.7	
	MCWB	24.7	25.5	25.6	22.6	20.5	17.8	17.0	17.4	18.2	20.0	19.3	23.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	28.7	29.0	28.2	26.3	24.2	21.2	21.8	23.3	24.8	26.1	26.9	27.7	
	2% MCWB	24.1	24.3	24.2	22.0	19.1	16.7	16.4	16.2	18.3	20.3	22.0	22.6	
	5% DB	27.8	27.8	27.1	25.2	22.7	20.2	20.4	21.6	23.5	24.7	25.8	26.8	
	5% MCWB	23.5	23.4	23.3	21.2	18.5	16.4	16.0	15.9	18.2	20.0	21.4	22.3	
	10% DB	27.0	27.0	26.1	24.3	21.7	19.4	19.2	20.2	22.2	23.7	24.8	26.0	
	10% MCWB	23.2	22.9	22.5	20.6	17.9	16.0	15.3	15.2	17.6	19.5	20.8	22.0	
	WB	26.1	26.7	26.4	24.5	22.3	20.0	19.9	19.5	21.2	23.0	23.7	25.1	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	27.9	28.4	28.7	25.7	24.4	20.7	21.6	23.4	24.0	25.0	26.7	27.6	
	2% WB	25.1	25.7	25.2	23.2	20.9	18.6	18.5	18.4	20.3	21.9	22.9	24.0	
	2% MCDB	27.3	27.2	26.8	25.0	22.4	19.9	20.1	21.0	22.8	24.1	25.5	26.0	
	5% WB	24.5	24.8	24.5	22.2	19.8	17.8	17.2	17.5	19.5	21.0	22.2	23.3	
	5% MCDB	26.6	26.7	26.1	24.5	21.6	19.2	19.2	20.1	21.9	23.3	24.5	25.4	
	10% WB	23.9	24.0	23.7	21.2	18.8	17.0	16.1	16.6	18.7	20.2	21.7	22.7	
	10% MCDB	25.8	25.9	25.4	23.6	21.0	18.5	18.2	19.3	21.2	22.6	23.8	24.8	
Mean Daily Temperature Range	MDBR	5.7	5.6	5.8	6.6	8.0	7.5	8.6	9.3	8.1	7.1	6.2	5.9	
	5% DB	6.9	6.8	7.0	8.0	9.2	8.3	9.9	11.1	9.6	9.1	7.7	7.3	
	MCWBR	3.3	3.2	3.7	4.4	5.3	5.2	6.0	6.5	5.7	5.0	4.0	3.8	
	5% WB	6.0	6.1	6.6	7.5	7.8	6.7	7.9	9.5	8.4	7.8	6.4	6.5	
	MCWBR	3.3	3.4	4.1	4.5	5.0	5.1	5.4	6.4	5.4	4.8	3.8	3.7	
Clear-Sky Solar Irradiance	taub	0.388	0.373	0.357	0.339	0.315	0.310	0.301	0.314	0.356	0.385	0.385	0.392	
	taud	2.512	2.565	2.605	2.630	2.655	2.655	2.666	2.610	2.488	2.436	2.484	2.485	
	Ebn at Noon	954	950	935	904	883	864	890	917	920	927	950	952	
	Edh at Noon	114	105	95	84	74	70	72	84	105	118	116	117	
All-Sky Solar Radiation	RadAvg	6.56	5.99	5.02	4.24	3.46	2.87	3.31	4.21	5.29	5.92	6.39	6.52	
	RadStd	0.52	0.46	0.29	0.21	0.21	0.17	0.26	0.29	0.37	0.51	0.54	0.60	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	+0.34	N/A	N/A	+0.64	N/A	N/A	N/A	-45	+161	+101

Nomenclature: See separate page